



ArcGIS: Automating and Enhancing Apps for Accessibility and Localization



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2026 ESRI USER CONFERENCE

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<https://link.esri.com/sessioncodes/mixhalotechpres>

Agenda

- Goal and purpose
- Process and architecture
- Accessibility in ArcGIS Apps
- Survey configuration
- Boosting accessibility
- Automation
- Further discussion

Goal and purpose

Project goals

Eastern gray squirrel morphs

- Map our furry friends
- Crowdsource support for different languages



Less than 1% of Eastern gray squirrels are melanistic or black.



Why accessibility and localization matter

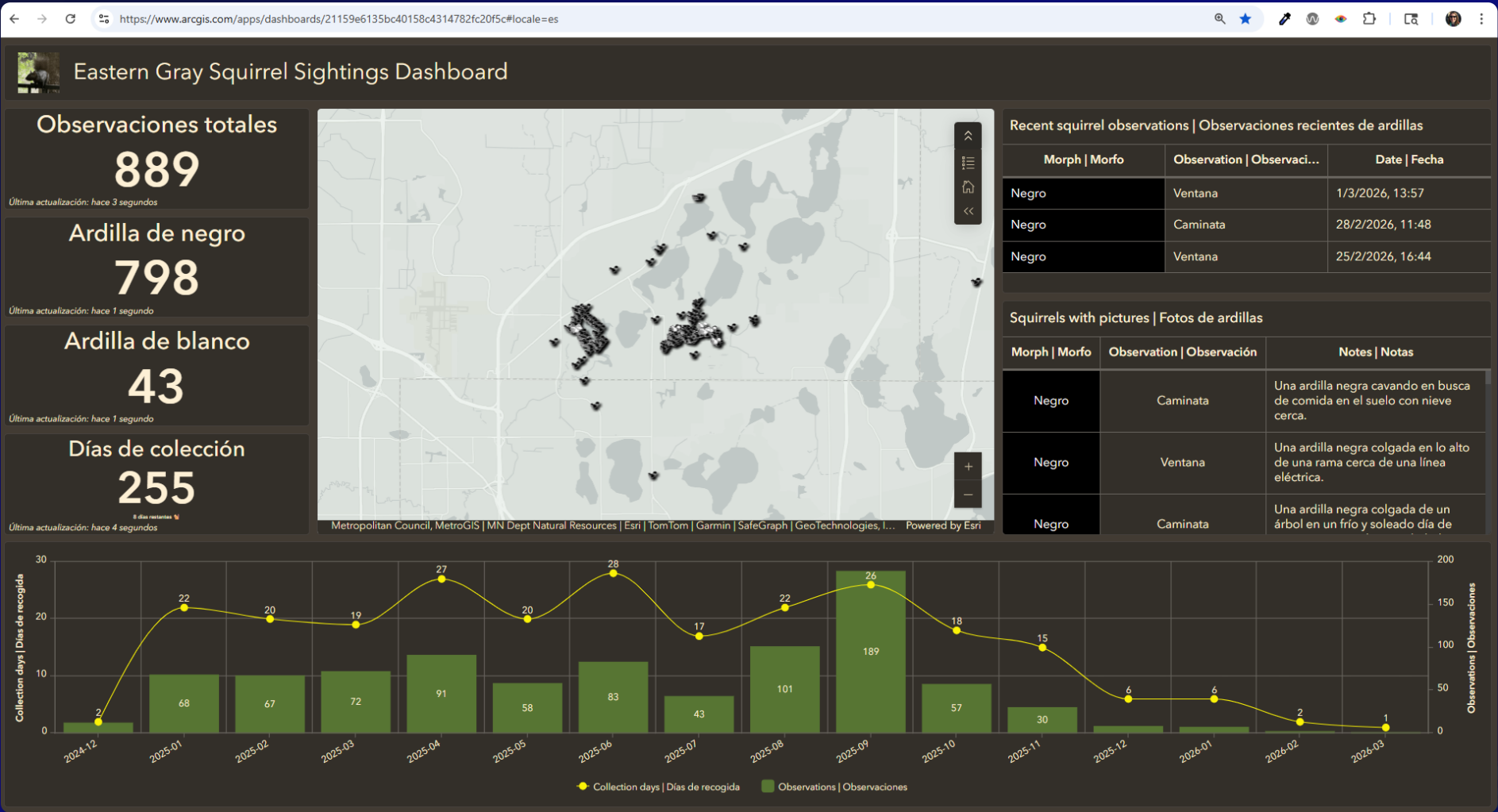
Design usable and inclusive experiences



- It is the right thing to do
- Remove barriers to access
 - Allow access regardless of ability
 - Offer greater usability
- Adapt to local and cultural standards
 - Provide an expected experience without altering underlying data
 - Date and number formatting
 - Offer the same content across languages
- Deliver the same content across languages

Squirrels dashboard

Support for more audiences



Process and architecture

Squirrel collection: Before

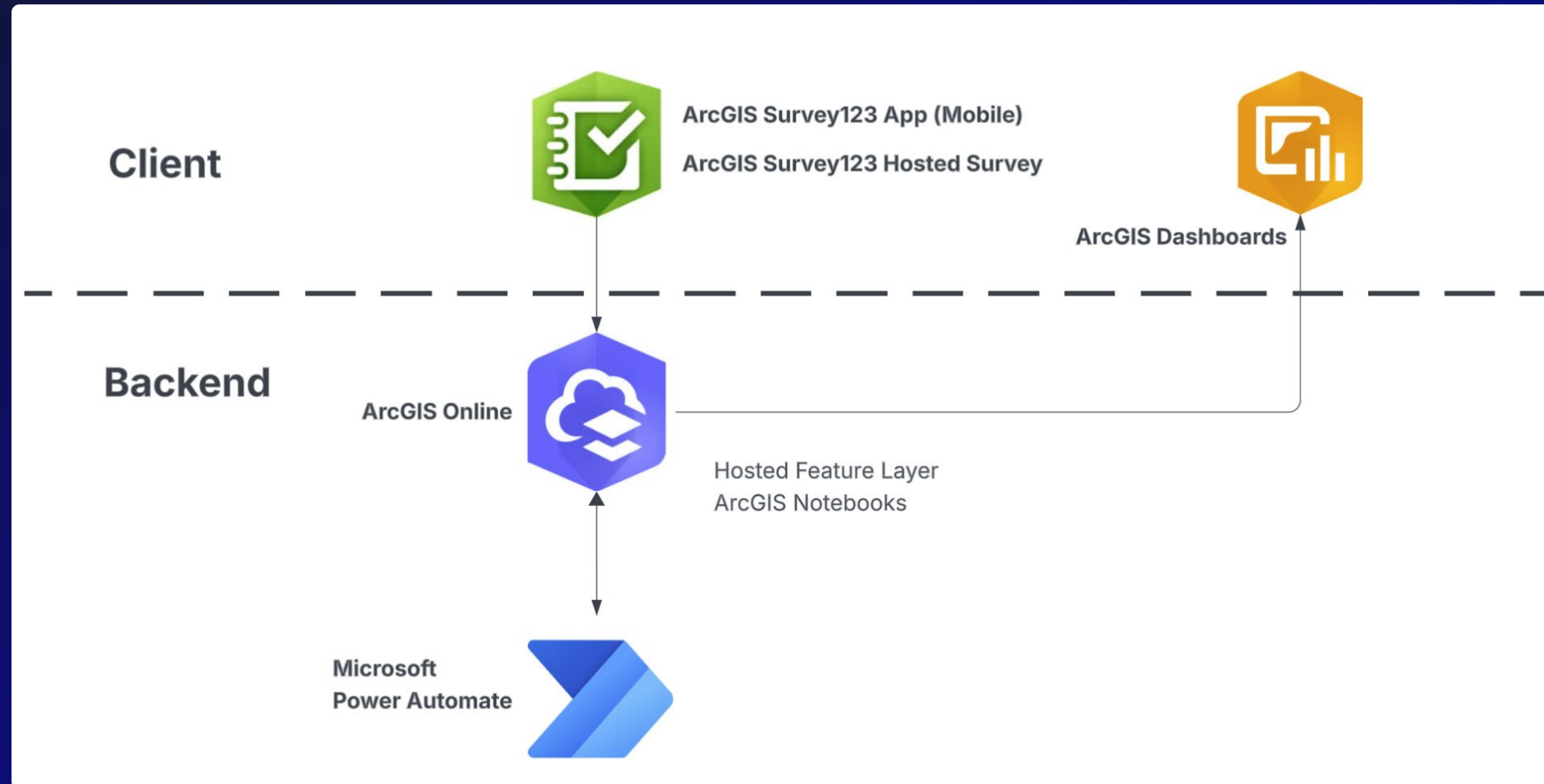
Tedious data entry ~30 minutes

- Data collection could not be collected in the field
 - Accuracy issues resulting in collection 20 minutes to days later
 - Convert images from .heic to .png
- Internationalization
 - Required manual use of Google's translator for each observation to ensure multiple languages were offered
- Accessibility
 - Follow-up to ensure each photos had its respective alt text



Updated architecture

End-to-end process ~2 minutes



Accessibility in ArcGIS apps

Data collection usability

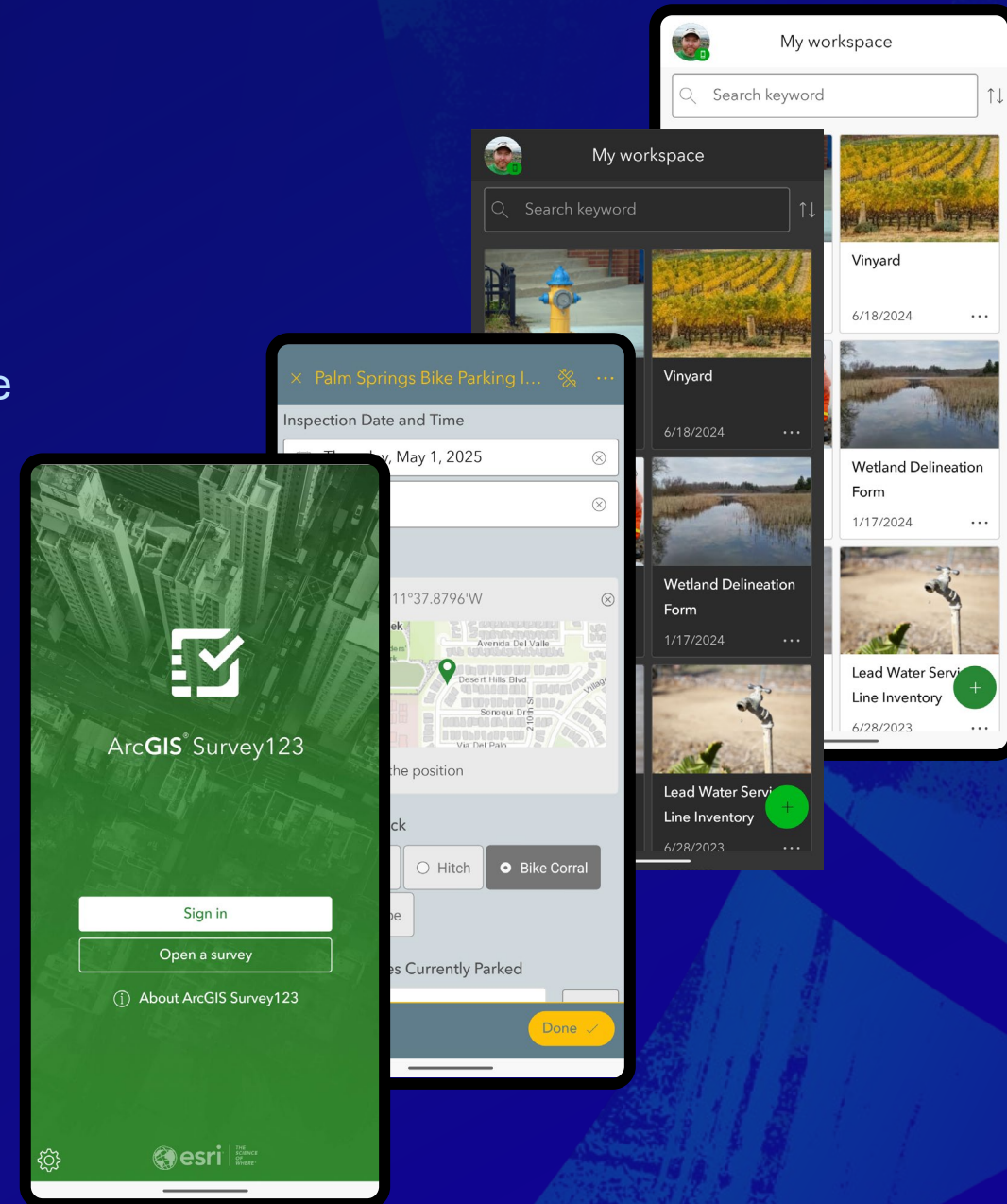
Designing maps for contrast, clarity, and context

- ArcGIS Survey123

- Structured from inputs with named role, state, and value
- Interaction with UI through assistive technologies
- Status communicated to assistive technologies
- Required field logic
 - Conditional note requirements
 - Photo attachments

- Usability experience

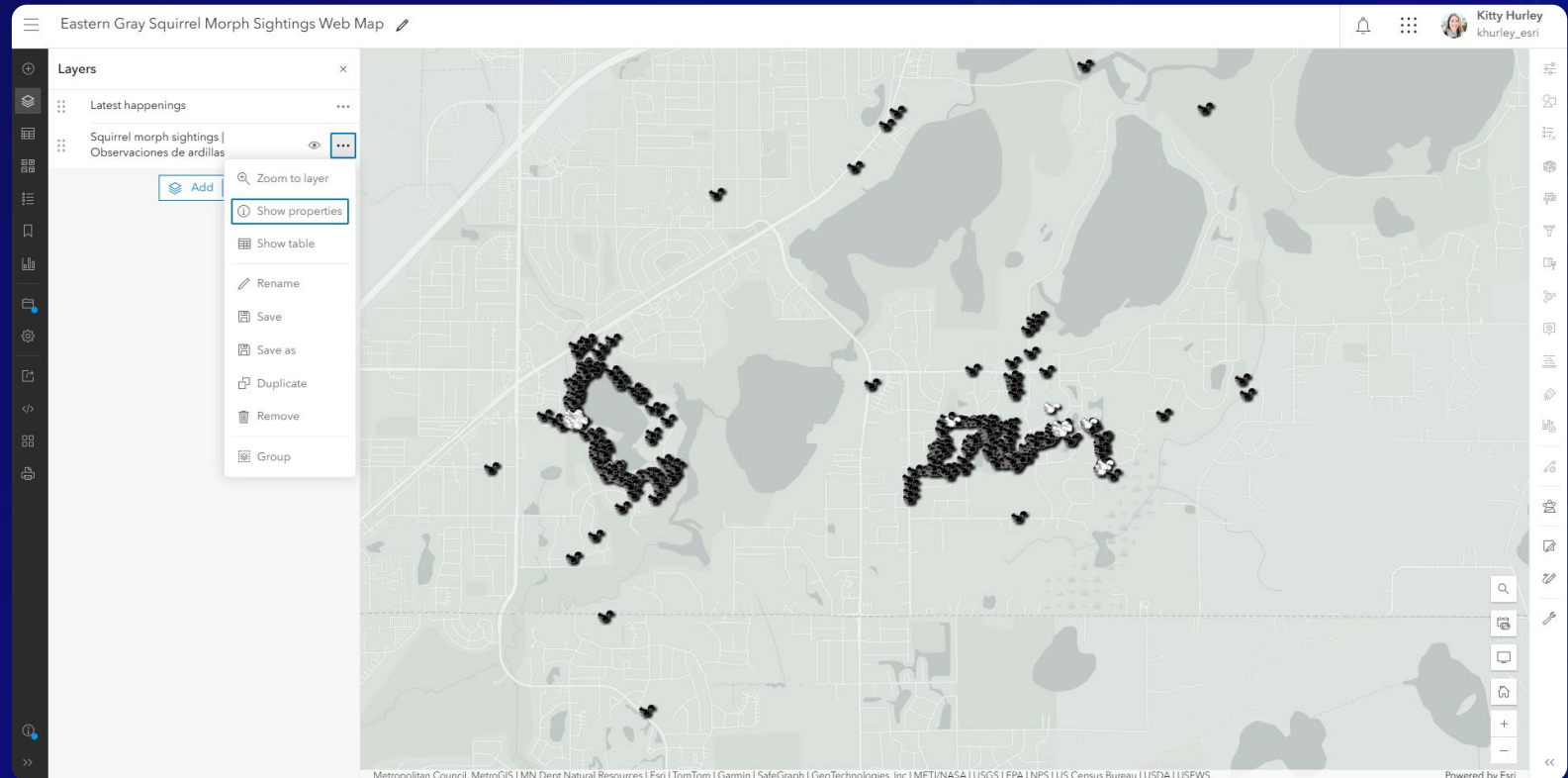
- Supports cognitive users
- Ease-of-use
- Greater access to individuals



Map usability

Using ArcGIS Map Viewer

- Keyboard navigation
- Element labels provide context
- Further customize solutions

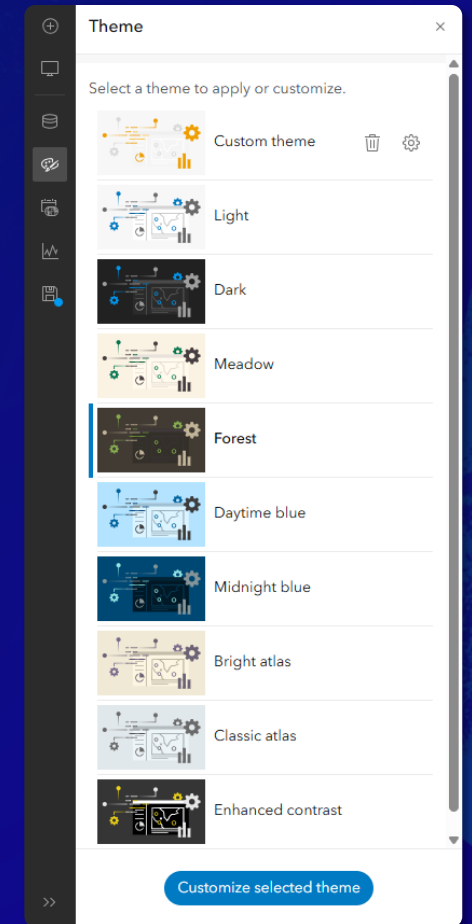


ArcGIS Map Viewer provides consistent keyboard navigation.

Accessibility with ArcGIS Dashboards

Inclusive analytics with adaptive interface design

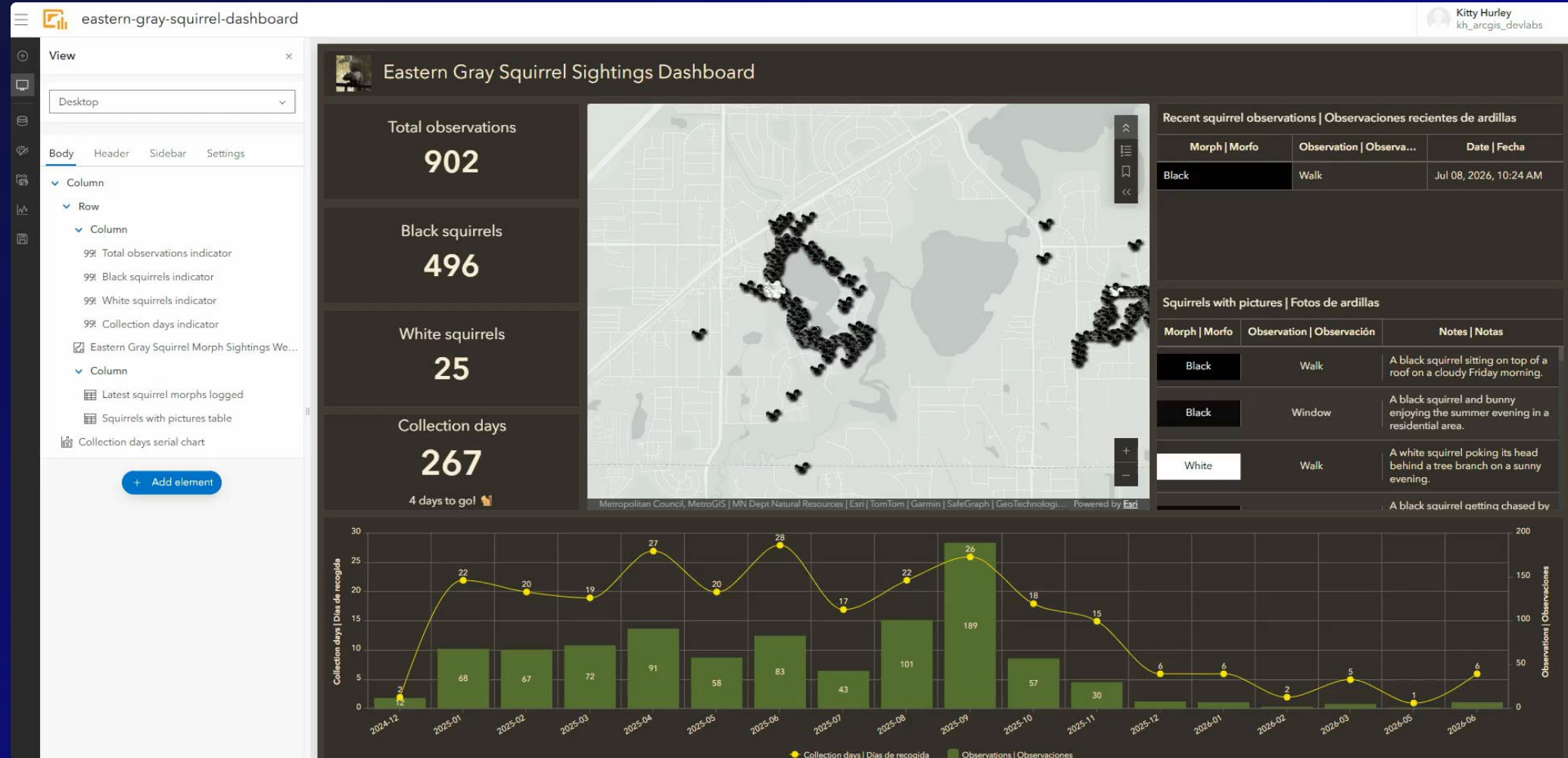
- Color theme contrast
- Element labeling
- Keyboard navigation
- Multiple visualization elements
- Supports locales based on default browser setting and through URL parameters
- Support for multi-locales for some Dashboard elements
 - Headings
 - Indicator elements using Arcade
- Desktop and mobile views



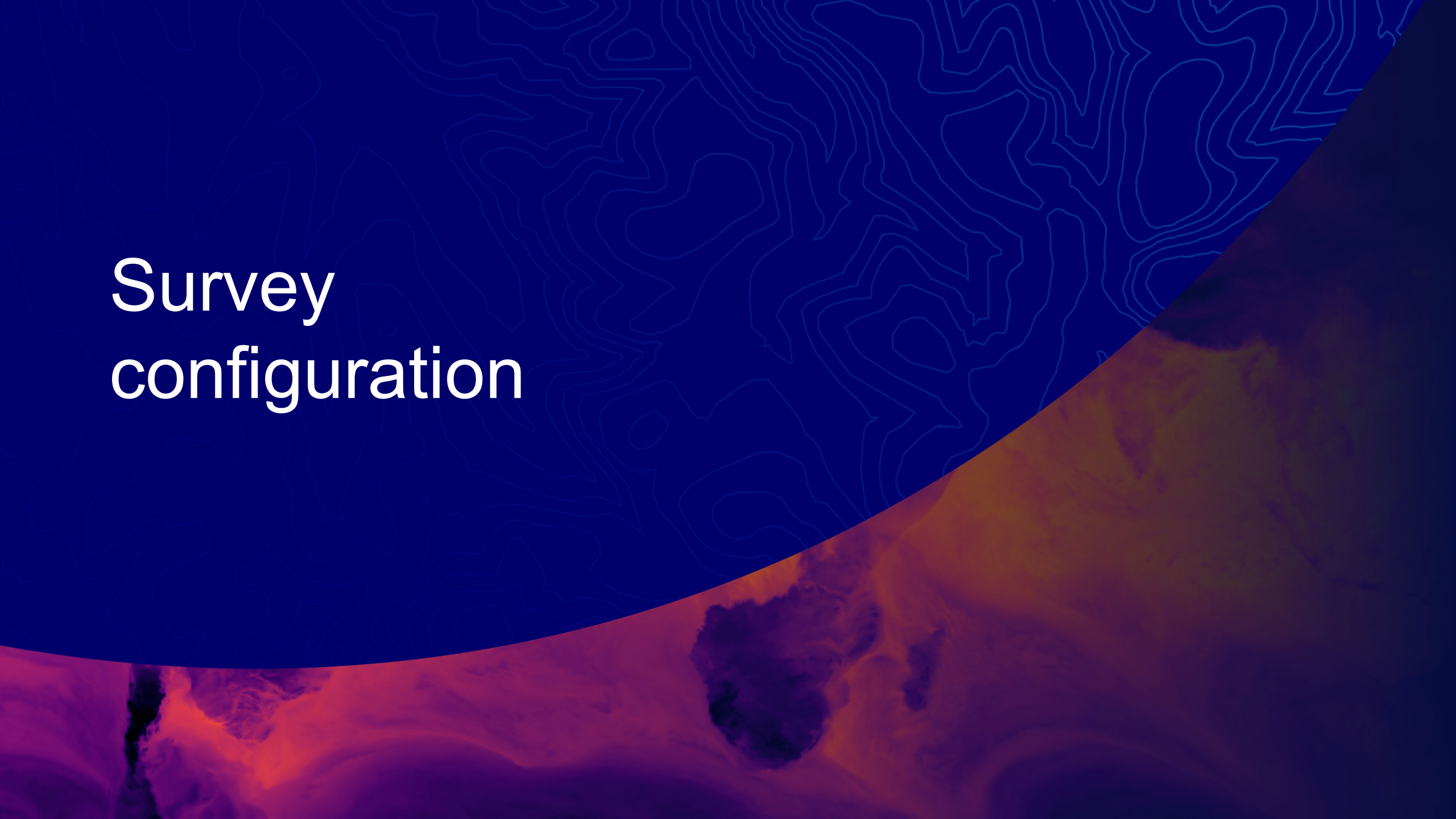
ArcGIS Dashboards color themes configuration options

Usability with ArcGIS Dashboards

Inclusive analytics with adaptive interface design



Survey configuration

The background of the slide is split into two main sections. The upper section is a solid dark blue color, overlaid with a white, intricate topographic map pattern consisting of many irregular, concentric lines. The lower section is a curved, abstract shape that transitions from a deep purple on the left to a bright orange on the right, with a dark, silhouetted landscape or cloud formation visible within the purple area.

Data collection

Accessibility starts at data entry

- Setup the configuration of survey form
- Ease of use for Survey123 app
- Crowd sourced collection
- ArcGIS Online
 - Hosted Feature Service
- Alternative product considerations
 - ArcGIS Field Maps

Black morphs have lower heat loss and basal metabolic rates during winter months.



Squirrel data collection

Survey123

- Morph Color
 - **Black** || **White**
- Observation type
 - **Walk** || **Window**
- Date and Time
- Photo (*optional*)
- Notes (*optional, unless a photo is provided*)

<

Squirrel Entry



Eastern Gray Squirrel morph sightings including black, or melanistic and white, or al...

Owner: khurley_esri
Created: 6/18/25 11:35 AM
Modified: 12/16/25 4:39 PM

Eastern gray squirrel morph sightings including black, or melanistic and white, or albino collection observations.

 Collect >

 Sent 471 >

Boosting accessibility

The background of the slide is split into two main horizontal sections. The top section is a solid dark blue color, overlaid with a white, intricate topographic map pattern consisting of many irregular, concentric lines. The bottom section is a curved, gradient area transitioning from a deep purple on the left to a bright orange on the right, with some darker, cloud-like shapes visible within the gradient.

Data visualization

Map configuration using Map Viewer

- Basemap
 - Consider contrast with data
 - Reference labels
- Symbology
 - Recent observations flashing
- Popups
 - Arcade expressions

Black morphs often act bolder and more aggressive around predators than their gray counterparts.



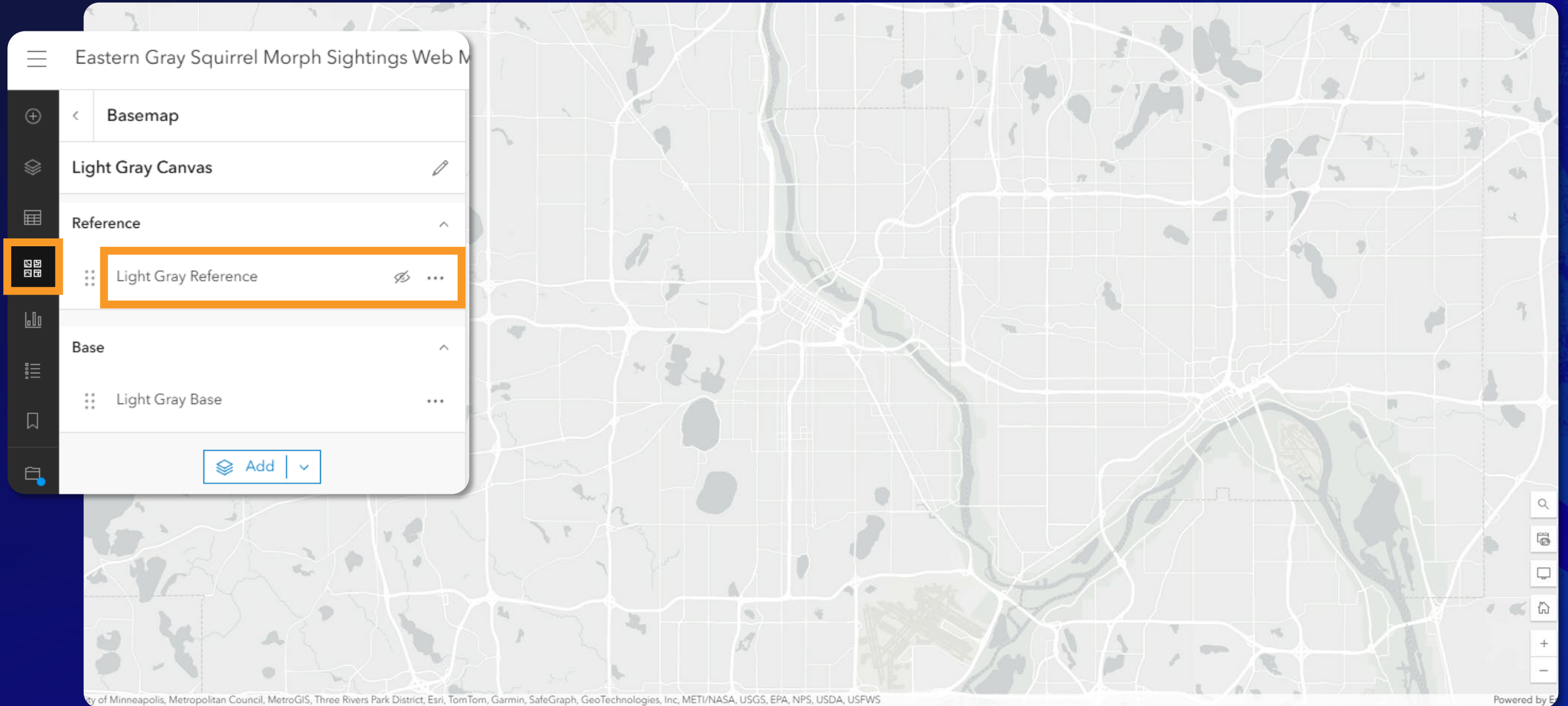
Map design considerations

Designing maps for contrast, clarity, and context

- Add “pop” to our point data
- Provide contrast between point data and base layers
- Create dynamic popups based on the data
- Offer support for locales
- Format dates for different locales
- Provide alternative text for our photos
- Focus on recent observations

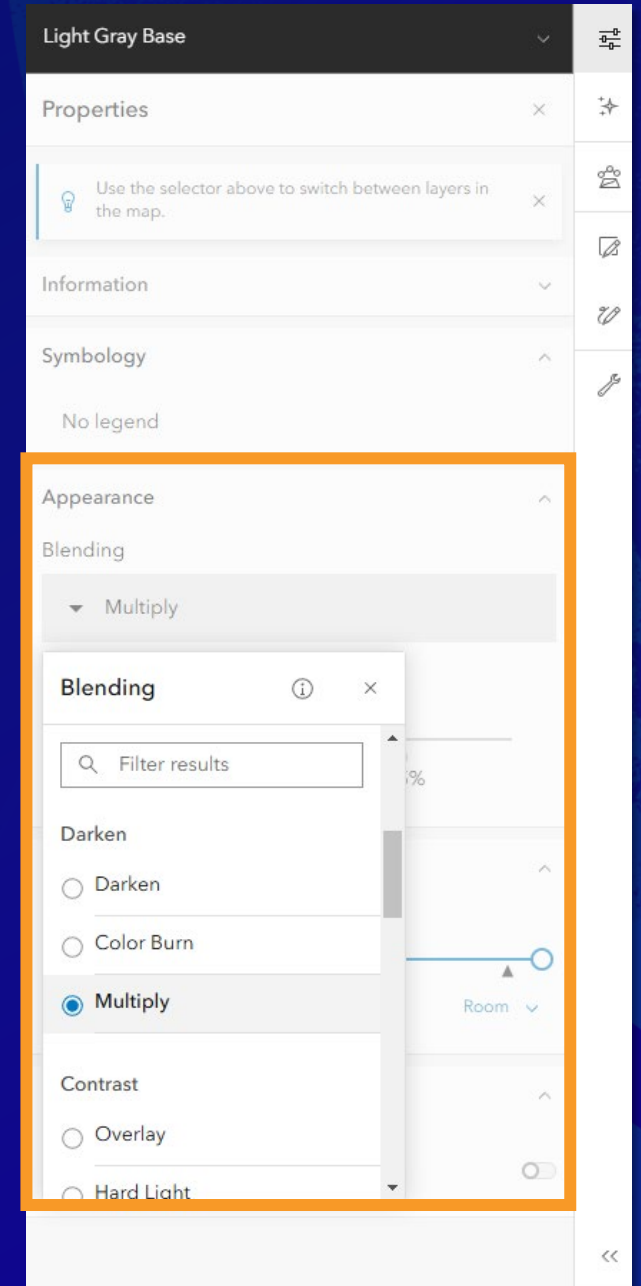
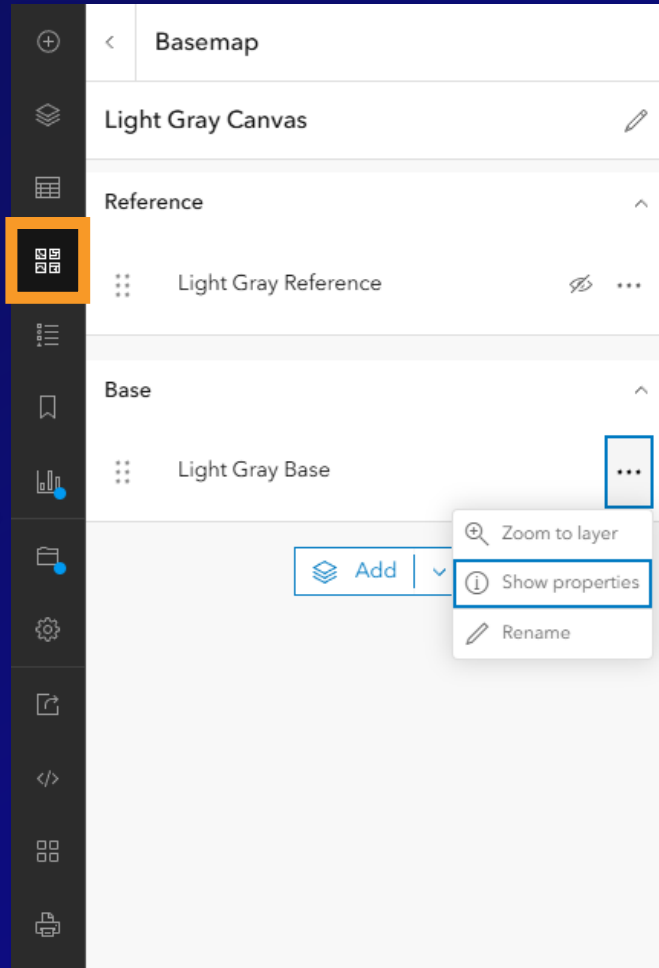
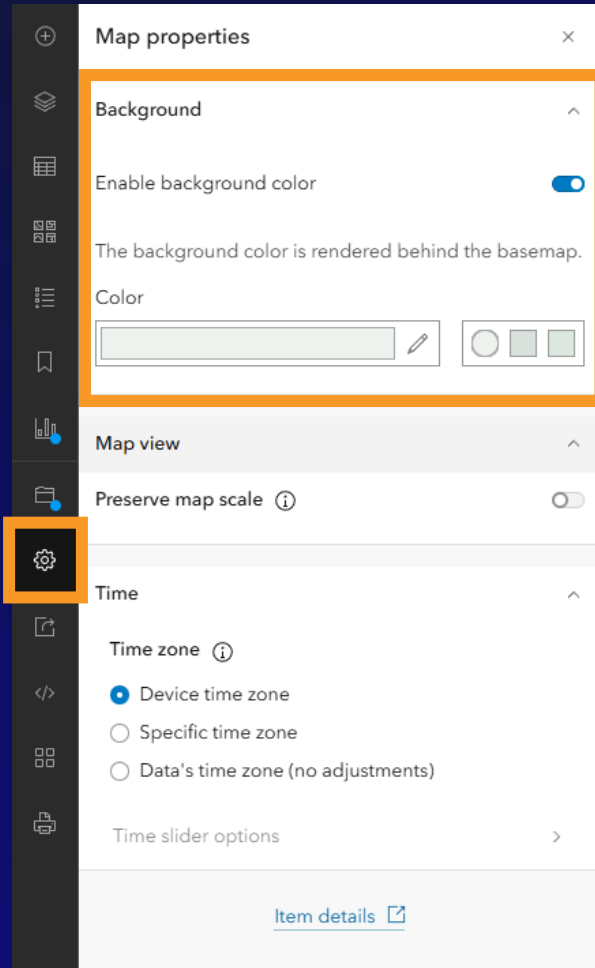


Basemap hierarchy: Reference



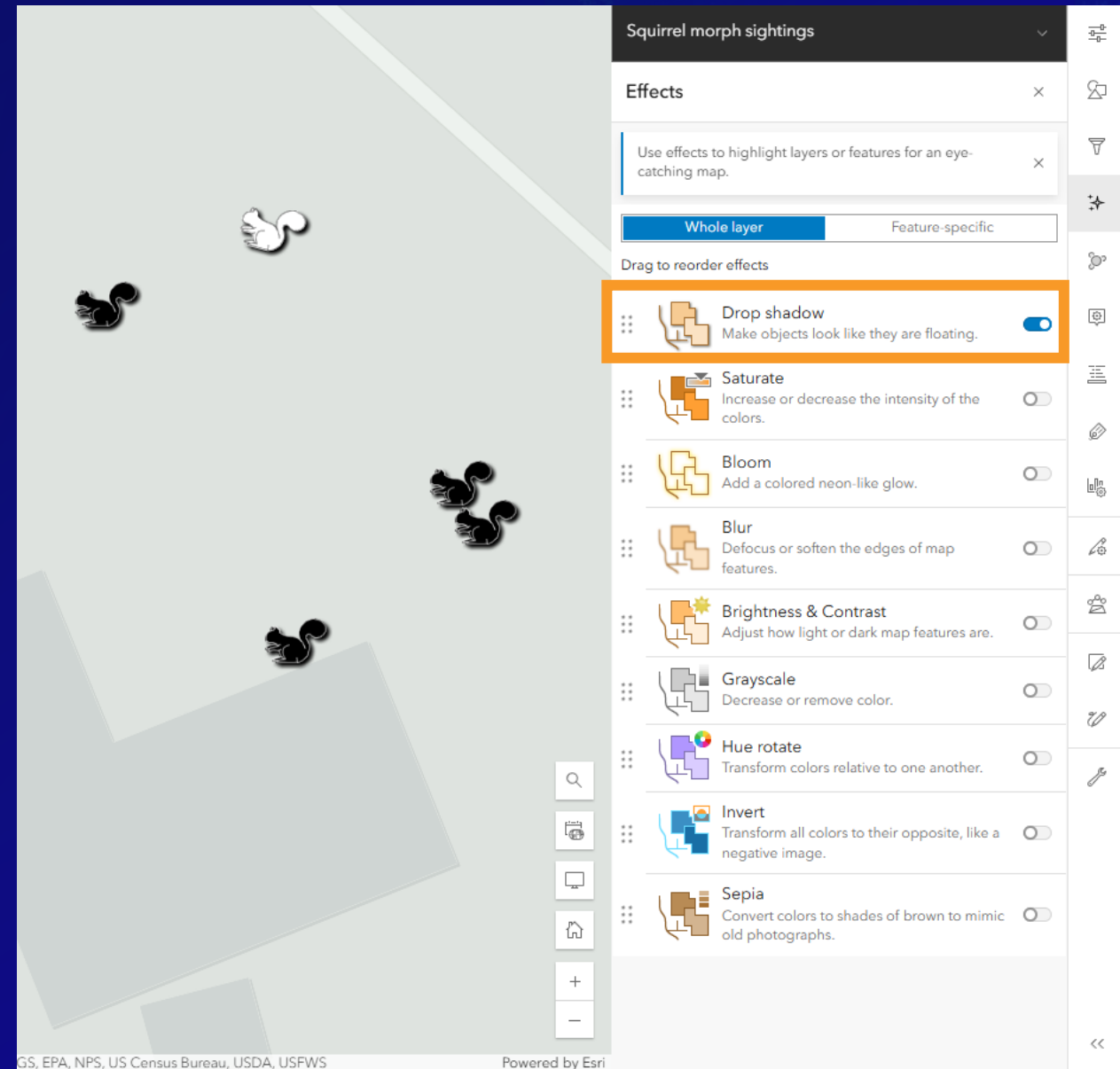
Basemap visualization

Basemap multiply blending



Add “pop” to the data

- Squirrel symbology
- Display features by value order
- Point symbol rotation
- Layer effects
 - Drop shadow



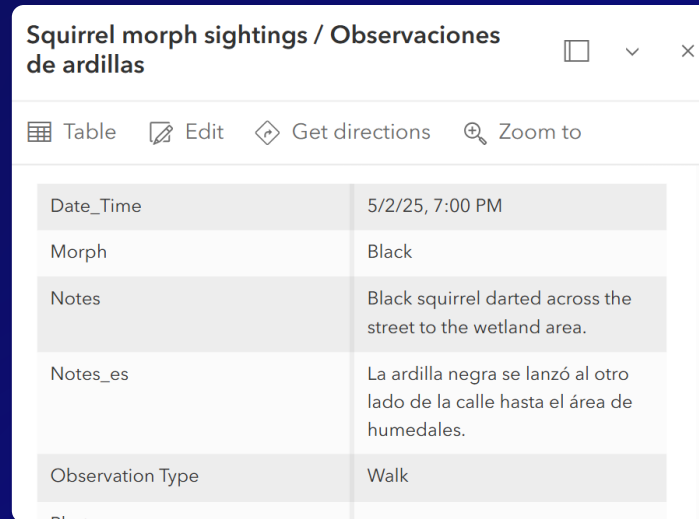
Visualizing the squirrel morphs



Accessible popups

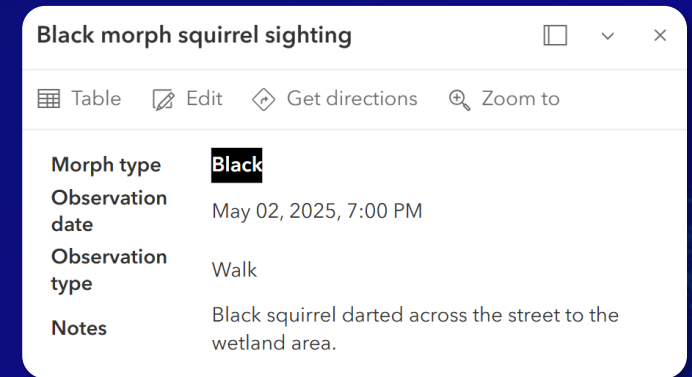
Providing structured context for assistive technologies

- Clear field labeling
- Use Arcade expressions
- Display data in table format
- Alt-text exposure
- Language-aware content



Date_Time	5/2/25, 7:00 PM
Morph	Black
Notes	Black squirrel darted across the street to the wetland area.
Notes_es	La ardilla negra se lanzó al otro lado de la calle hasta el área de humedales.
Observation Type	Walk

Default popup (English)



Morph type	Black
Observation date	May 02, 2025, 7:00 PM
Observation type	Walk
Notes	Black squirrel darted across the street to the wetland area.

Configured popup (English)

Dynamic popup title

ArcGIS MapViewer and Arcade functions

```
// Get the locale
var locale = GetEnvironment().locale;

// Set the popup title based on the locale
if (locale == "es") {
  return IIF(
    $feature.Morph_type == "Black",
    "Avistamiento de ardilla de morfo negro",
    "Avistamiento de ardilla de morfo blanco"
  );
} else {
  return `${$feature.Morph_type} morph squirrel sighting`;
}
```

Eastern gray squirrels
fake-bury food when
watched to prevent
theft.



Create dynamic popup content

ArcGIS MapViewer and Arcade

```
// Dynamically generate squirrel color
var squirrelColor = IIF($feature.Morph_type == "Black", "white", "black");
var backgroundColor = IIF(squirrelColor == "black", "#CCC", "black");

// Get the locale
var locale = GetEnvironment().locale;

if (locale == "es") {
  var morphColor = IIF($feature.Morph_type == "Black", "Negro", "Blanco");
  var dateValue = Text($feature.Date_Time, "D/M/YYYY,");
  var timeValue = Text($feature.Date_Time, "HH:mm");
  var observationType = IIF($feature.Observation_Type == "Walk", "Caminata",
"Ventana");
  var notesValue = $feature.Notes_es;
} else {
  var morphColor = $feature.Morph_type;
  var dateValue = Text($feature.Date_Time, "MMM DD, YYYY, ");
  var timeValue = Text($feature.Date_Time, " h:mm A");
  var observationType = $feature.Observation_Type;
  var notesValue = $feature.Notes;
}
```

Values dynamically generated based on the locale.

```
// Display notes only when provided and without a photo
if (HasValue($feature, "Notes") && IsEmpty($feature.Photo_URL)) {
  var squirrelNotes = `|
    <th>${notesHeading}</th>
    <td>${notesValue}</td>
  </tr>`;
} else {
  squirrelNotes = "";
}

// Display a photo and include the note as the alt tag
if (HasValue($feature, "Photo_URL")) {
  var squirrelPhoto = `|
    <th>${photoHeading}</th>
    <td>
      
    </td>
  </tr>`;
} else {
  var squirrelPhoto = "";
}
|  |

|  |

```

Dynamically populate an alt-text for photos, where present.

Store dynamic popup content in a table

ArcGIS MapViewer and Arcade

```
// Return the content in a table format for better ally and i18n support
return {
  type: "text",
  text: `<table style="text-align:left">
    <tr>
      <th>${morphHeading}</th>
      <td>
        <span style="background-color:${backgroundColor};
          color:${squirrelColor}">
          <b>${morphColor}</b>
        </span>
      </td>
    <tr>
      <th>${dateHeading}</th>
      <td>${dateValue} ${timeValue}</td>
    </tr>
    <tr>
      <th>${observationHeading}</th>
      <td>${observationType}</td>
    </tr>
    ${squirrelNotes}
    ${squirrelPhoto}
  </table>`,
};
```

While white fur can be more visible to predators, snowy environments provide camouflage.



Dynamic popups by locales

English and Spanish



Black morph squirrel sighting

Table Edit Get directions Zoom to

Morph type **Black**
Observation date May 02, 2025, 7:00 PM
Observation type Walk
Notes Black squirrel darted across the street to the wetland area.



Avistamiento de ardilla de morfo negro

Tabla Editar Obtener indicaciones Acercar

Tipo de morfo **Negro**
Fecha de observación 2/5/2025, 19:00
Tipo de observación Caminata
Notas La ardilla negra se lanzó al otro lado de la calle hasta el área de humedales.

White morph squirrel sighting

Table Edit Get directions Zoom to

Morph type **White**
Observation date Jul 07, 2025, 6:33 PM
Observation type Walk

Photo


Cookie getting some snacks on a hot summer evening.

Avistamiento de ardilla de morfo blanco

Tabla Editar Obtener indicaciones Acercar

Tipo de morfo **Blanco**
Fecha de observación 7/7/2025, 18:33
Tipo de observación Caminata

Foto


Cookie recibe algunos bocadillos en una calurosa noche de verano.

Popups configured to dynamically update to a locale and the data collection.

Localization without data duplication

Use browser locale and Arcade to adapt dynamically

- English and Spanish support
- Browser-based locale detection
- URL parameter overrides
- Date and number formatting

```
// Get the locale
var locale = GetEnvironment().locale;

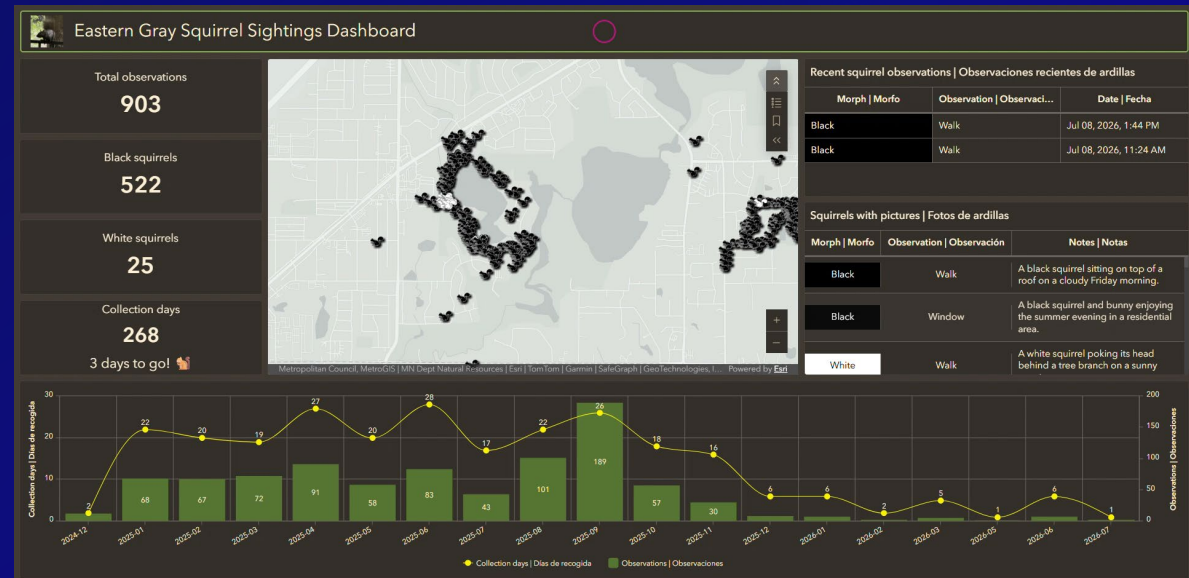
// Set the title based on the locale
if (locale == "es") {
  var squirrelText = "Ardilla de blanco";
} else {
  var squirrelText = "White squirrels";
}

return {
  topText: `${squirrelText}`,
  topTextMaxSize: "medium",
  middleText: $datapoint.COUNT_OBJECTID,
  middleTextMaxSize: "large",
};
```

The White morph indicator's Arcade expression in Dashboards, which adjusts to locales.

Keyboard navigation

- Support consistent navigation
- Consider element purpose
- Keep layouts intuitive



Assistive technology support

ArcGIS Dashboards



Serial chart

Data

Chart

Category axis

Value axis

Guides

Series

General

Actions

Accessibility

Accessibility options

Screen reader

Accessible name

Number of squirrel morph observations and days of collec...

Collection days | Días de recogida

Observations | Observaciones

Data table

MonthYear	OM_Count	CDM_UniqueDayCount	FID
2024-12	12	2	0
2025-01	68	22	1
2025-02	67	20	2
2025-03	72	19	3
2025-04	91	27	4
2025-05	58	20	5
2025-06	83	28	6

Showing 16 features

Cancel

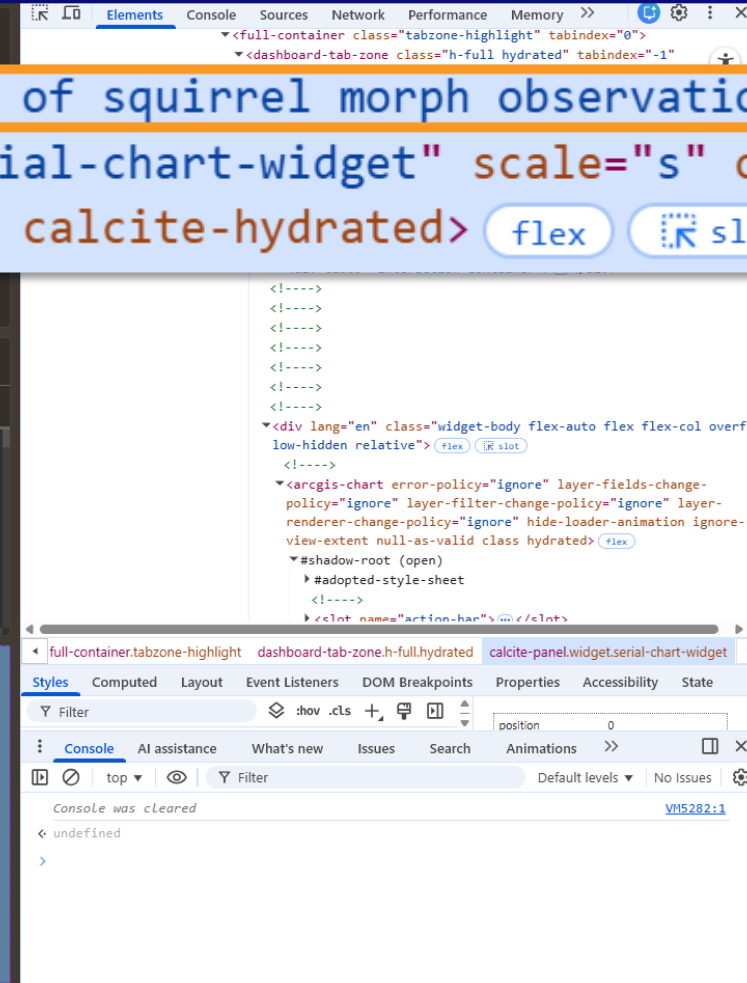
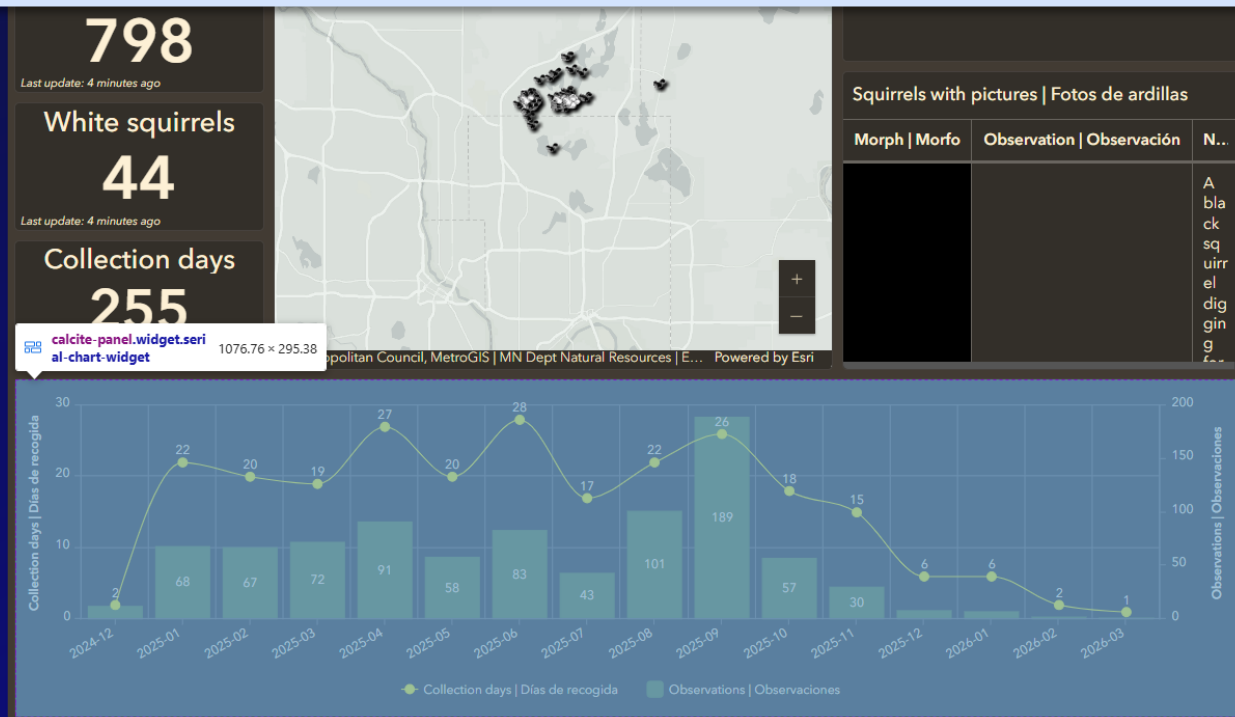
Done

Assistive technology support

ArcGIS Dashboards

Eastern Gray Squirrel Sightings Dashboard

```
<calcite-panel role="region" aria-label="Number of squirrel morph observations and days of collection per month" class="widget serial-chart-widget" scale="s" overlay-positioning="fixed" menu-placement="bottom-end" calcite-hydrated> flex slot
```



ARIA label best practices

- Describe the purpose of the element
- Use informative, plain language
 - Write for understanding, not for search engine optimization (SEO)

Eastern Gray Squirrel Morph Sightings Web Map

Map

Settings

General

Header

Title

{ } Eastern gray squirrel observations

More information

Edit

Enable focus mode

Header text color

Cancel Done

Eastern Gray Squirrel Morph Sightings Web Map

Map

Settings

Accessibility options

Screen reader

Accessible name

Eastern gray squirrel sightings interactive map with recent sightings highlighted.

Cancel Done

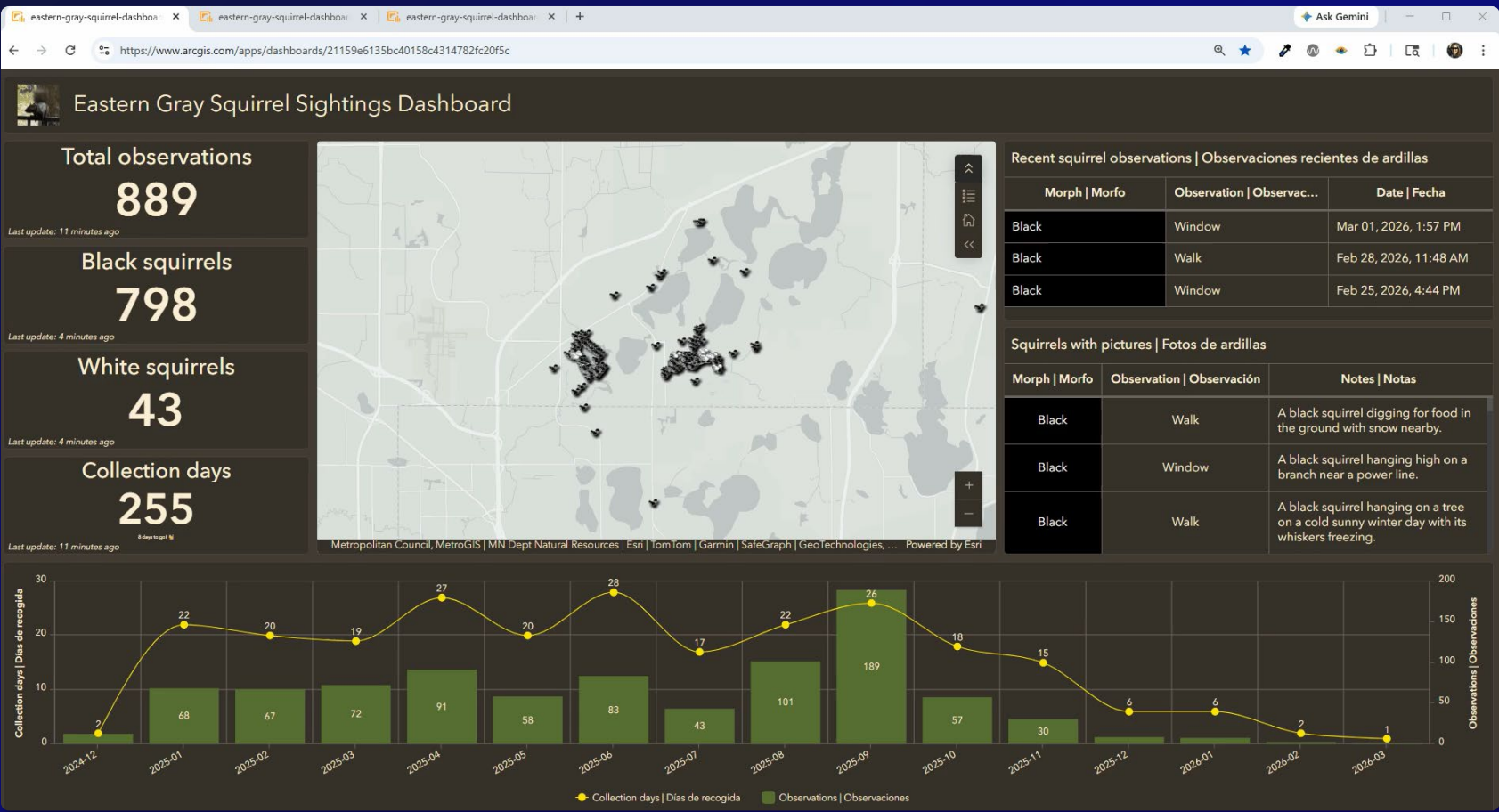
Use plain language

- Use **clear and concise** language
- **Avoid unnecessary** words and acronyms
- **Break out content** into lists and bullet points
- Use **present tense**
- **Provide labels and instructions** that describe the user's next action
- **Focus on clarity** over technical terminology

Use	Avoid
Do	Implement
Many	Numerous
Before	Prior to
Finish	Finalize
Recommend	Avoid
About	Approximately

Squirrel visualization and analysis

ArcGIS Dashboards

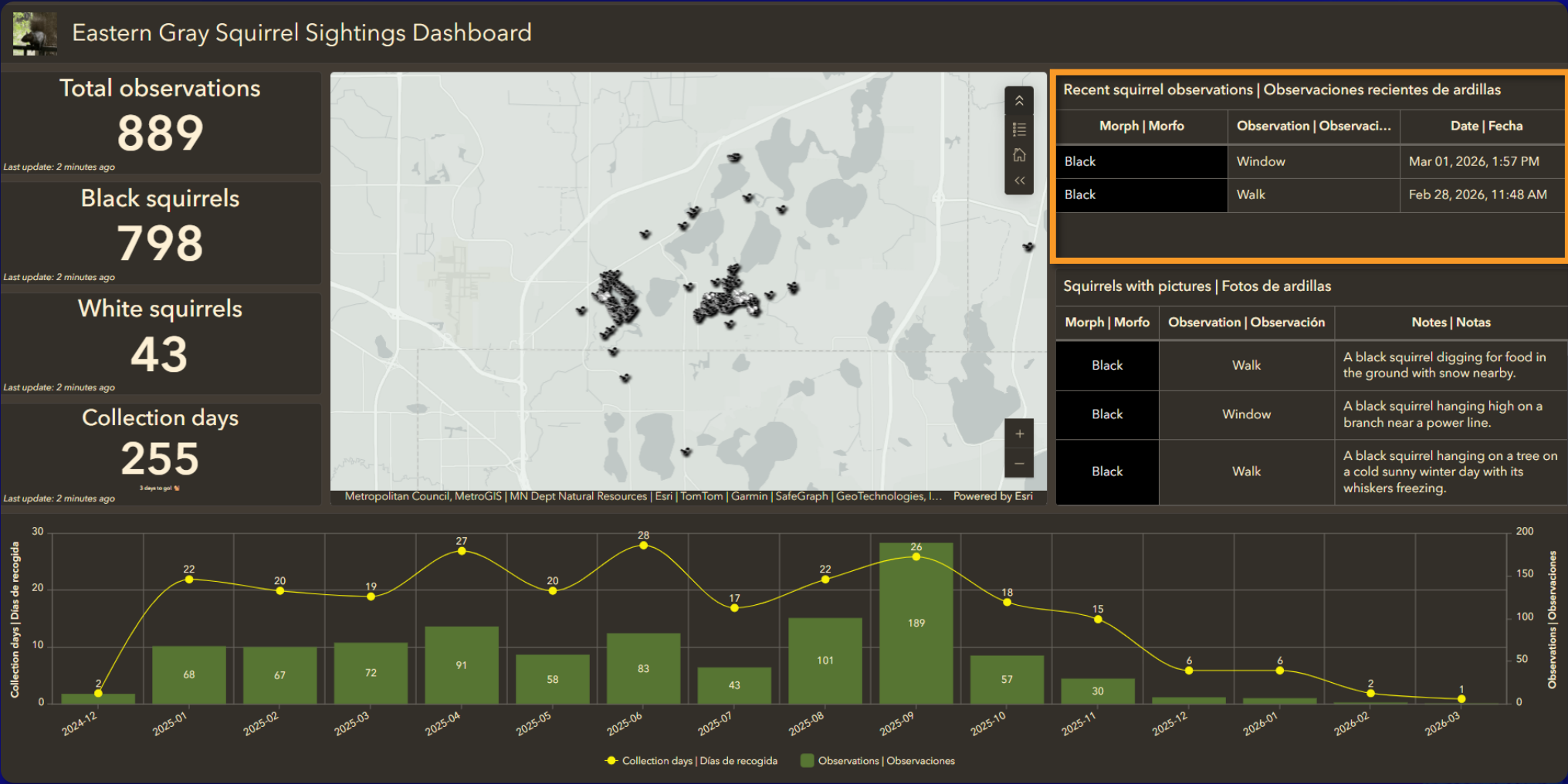


Black squirrels have higher concentrations of dark pigments caused by a genetic mutation.



Recent squirrels table

ArcGIS Dashboards



Recent squirrels table

ArcGIS Dashboards

```
// Get the locale
var locale = GetEnvironment().locale;

if (locale == "es") {
  var squirrelMorphColor = IIF($datapoint.Morph_type == "Black", "Negro", "Blanco");
  var observationText = IIF($datapoint.Observation_Type == "Walk", "Caminata", "Ventana");
  var dateValue = Text($datapoint.Date_Time, "D/M/YYYY,");
  var timeValue = Text($datapoint.Date_Time, "HH:mm");
} else {
  var squirrelMorphColor = $datapoint.Morph_type;
  var observationText = $datapoint.Observation_Type;
  var dateValue = Text($datapoint.Date_Time, "MMM DD, YYYY, ");
  var timeValue = Text($datapoint.Date_Time, " h:mm A");
}

// Colors for squirrels text
var squirrelBackgroundColor = $datapoint.Morph_type;
var squirrelTextColor = IIF(squirrelBackgroundColor == "Black", "#fff", "#000");
```

```
return {
  cells: {
    Morph_type: {
      displayText: `${squirrelMorphColor}`,
      textColor: `${squirrelTextColor}`,
      backgroundColor: `${squirrelBackgroundColor}`,
      textAlign: "left",
    },

    Observation_Type: {
      displayText: `${observationText}`,
      textAlign: "left",
    },

    Date_Time: {
      displayText: Text(`${dateValue} ${timeValue}`),
      textAlign: "left",
    },
  },
};
```

White morphs make up less than 1% of Eastern gray squirrels.





Demo: Let's log a squirrel

[English](#) || [Spanish](#)

Automation

The background of the slide is split into two main sections. The upper section is a solid blue area containing a white, intricate topographic map pattern of contour lines. A large, curved, abstract shape in shades of purple and orange separates this blue area from the lower section. The lower section features a dark, atmospheric image of a landscape with a bright, glowing orange light source, possibly a sunset or sunrise, creating a dramatic silhouette effect.

Event-driven automation

From feature creation to multilingual processing in seconds

- Remove barriers to access
- Support assistive technologies
- Adapt to cultural expectations
- Deliver the same content across languages

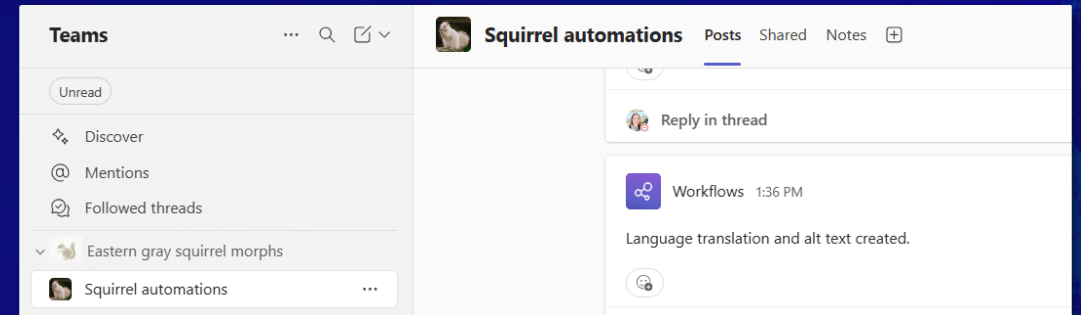
White squirrel morphs have distinct dark eyes, unlike albino squirrels, which lack pigment and have pink or red eyes.



Webhook

Event receiver and notification

- Feature Service Webhook
 - An event is initiated when a new entry is recorded
- Microsoft Power Automate orchestration
 - Listens to the creation of the event
 - Calls the ArcGIS Notebook
 - Delivers a message to a Microsoft Teams channel
- Microsoft Teams
 - Notifies owners that the entry has been successfully translated and made accessible



Processing with ArcGIS Notebooks

Automating translation and alt-text enforcement

- Google Translate API
- Attribute updates
- Alt-text validation
- Locale formatting

Ontario, Canada has some of the highest concentrations of black morphs at 67% of the squirrel population.



Automation options



Windows scheduled task

- Call notebook via web tool



ArcGIS Online scheduled task

- Call notebook
- Credit based consumption model



Pros

- Remove MS Power Automate from the equation



Cons

- Lose the ability of near-real time processing new squirrel entries

Further discussion

Design considerations

- Dashboards table widget support
- Webhooks not supported in Survey123 Next Gen
- Information passed from survey to webhook/MS Power Automate needs to be simple
- Why MS Power Automate
- Vetting translated alt-texts

White morphs thrive with humans present, such as urban parks and suburban cities.



A reusable architecture blueprint

Applying this pattern beyond squirrels

- This is an event-driven, accessibility-first architecture pattern
 - Environmental monitoring
 - Public health
 - Utility inspections
 - Citizen science
- How to fully automate dashboard based on particular locale
 - Notebook automation

Building accessibility into architecture

Inclusive GIS is better GIS

- Consider accessibility throughout the process
- Do not rely on manual translations
- Automate responsibly
- Design for everyone

About 1 in 10,000
Eastern gray squirrels
are black morphs.



Resources

- GitHub repo:
<https://esriurl.com/squirrels>
- Using Arcade to translate popups:
<https://esriurl.com/arcade-translate-popups>
- Five under-appreciated Arcade expressions:
<https://esriurl.com/5-arcade-expressions>

The screenshot shows the GitHub repository page for `geospatialem/eastern-gray-squirrels`. The repository is public and has 12 commits. The main branch is selected. The repository description is "Eastern Gray Squirrels 🐿️". The repository is for data and analysis on the Eastern gray squirrel. The project uses the ArcGIS technology stack to explore patterns in squirrel distribution and color morph variation. The repository includes a README, a LICENSE, and a .gitignore file. The repository is licensed under Apache-2.0. The repository has 0 stars, 0 forks, and 0 watchers. The repository has 3 contributors: geospatialem (Kitty Hurley), kgonzago (Kevin Gonzago), and Copilot. The repository has 2 languages: JavaScript (68.5%) and Jupyter Notebook (31.5%).

File	Commit	Time
arcade	feat(arcade): update collection date (#10)	7 months ago
python	docs(python): add squirrel photo + notes translation automat...	9 months ago
.gitignore	chore: add gitignore (#6)	9 months ago
LICENSE	Initial commit	9 months ago
README.md	Update README.md (#12)	3 months ago

Eastern Gray Squirrels 🐿️

A repository for data and analysis on the Eastern gray squirrel.

Overview

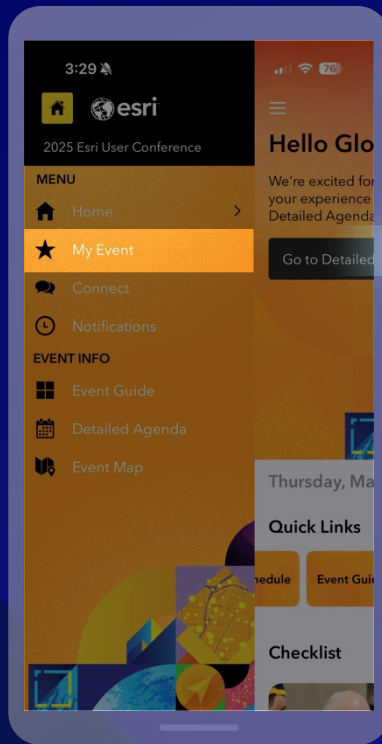
This repository collects and analyzes data on the **Eastern gray squirrel**. The project uses the ArcGIS technology stack to explore patterns in squirrel distribution and color morph variation.

Features

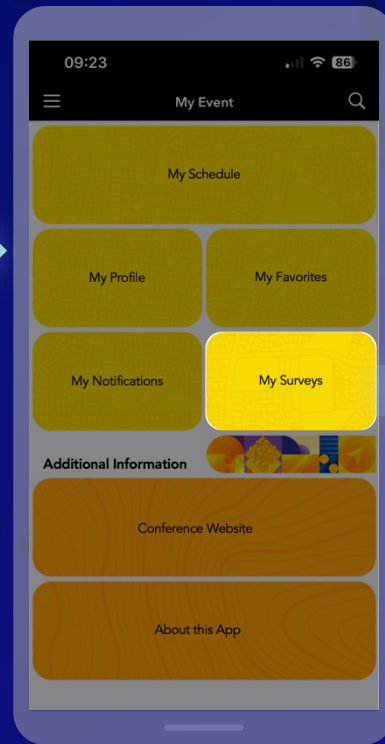
- Data preprocessing scripts for squirrel observation
- Exploratory data analysis including:
 - Distribution of sightings
 - Analysis of fur color variation
- Visualization to explore squirrel data

Please share your feedback in the app

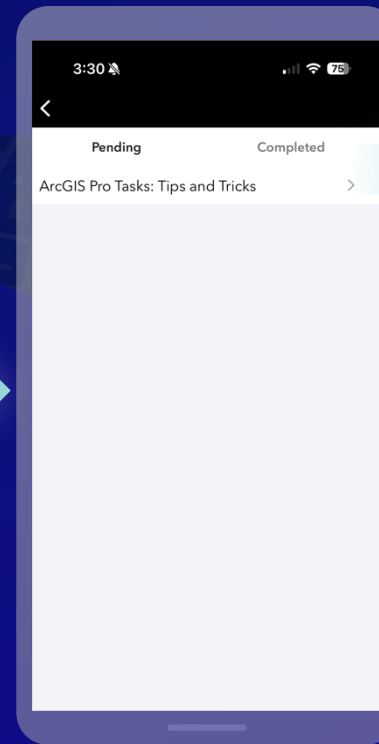
Select “My Event” in the event app main menu



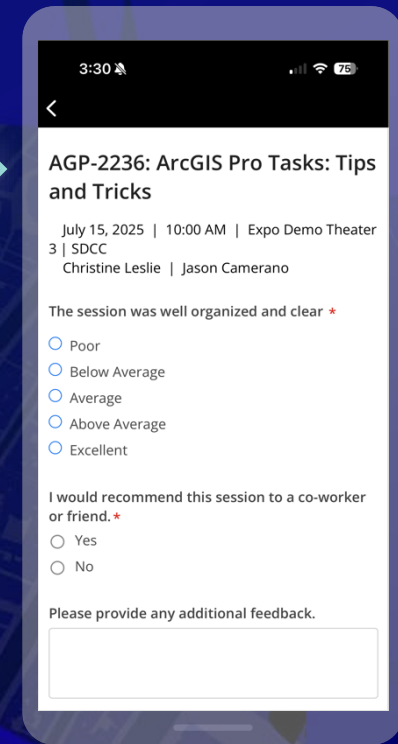
Select “My Surveys”



Select the desired survey



Complete the survey



Accessibility sessions at UC

<https://links.esri.com/a11y-uc-26>

» Tuesday, July 15

Accessibility Essentials for GIS and Mapping

8:30am Ballroom 20 A

1:00pm Room 15 B

ArcGIS Pro: Accessibility Features, Tips, and Tricks

1:00 pm Expo Theater 10



ArcGIS: Automating and Enhancing Apps for Accessibility and Localization

2:30 pm Room 17 A

» Wednesday, July 16

ArcGIS Online: Creating Accessible Content

10:00 am Expo Theater 14

Web Apps: Accessibility Features, Tips, and Tricks

2:30 pm Expo Theater 10

» Thursday, July 17

Web Apps: Accessibility Features, Tips, and Tricks

1:00 pm Expo Theater 8

Accessibility at UC Showcase





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SCIENCE
OF
WHERE[®]